

LineDiff: Semi-generative Line-drawing Frame Interpolation

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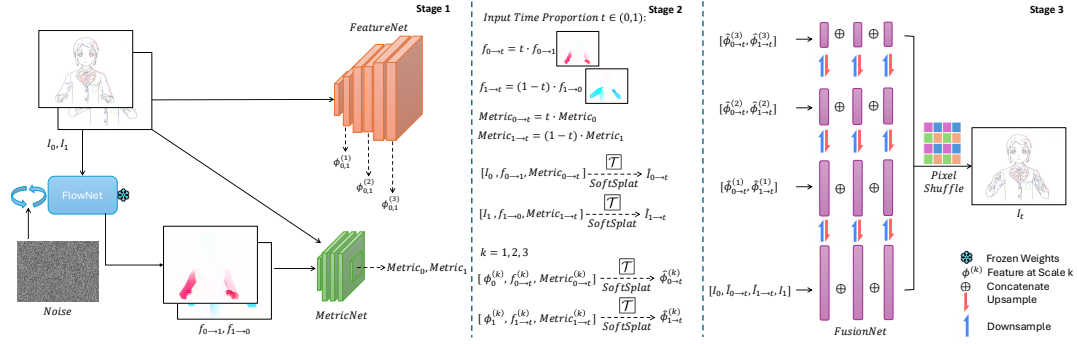


Figure 1: The overall workflow of LineDiff.

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1 Introduction

Automatic in-betweening of hand-drawn animation is difficult because line art lacks texture, color, and shading, making motion estimation challenging for conventional interpolation. Existing frame interpolation solutions fall into two main categories: static and generative. Static methods use optical flow and deterministic warping, performing well for small motions but failing with large displacements or missing correspondences. Generative methods introduce stochastic sampling, yielding creative, high-quality results but sometimes failing on simple cases.

We suggest a hybrid approach: using generative models not for full-frame synthesis, but to predict plausible motion flows. These flows then guide deterministic warping, combining the reliability of static methods with the adaptability of generative ones.

2 Methodology

We propose LineDiff, a generative in-between line-drawing interpolation network. The overview of LineDiff is shown in Figure 1. Given two input frames (I_0, I_1), the optical flows $f_{0→1}$ and $f_{1→0}$ between I_0 and I_1 in both directions are estimated through the FlowNet module. The foundation of FlowNet relies on FlowDiffuser[Luo

et al. 2024], a novel optical flow generative model that computes bidirectional optical flow between input frames. We fine-tuned a pre-trained FlowDiffuser model[Luo et al. 2024] on AnimeRun[Siyao et al. 2022] using contour and flow categories for estimating forward and backward motion.

While optical flow provides motion guidance, it may contain inaccuracies in occluded or motion-blurred regions. To mitigate such errors, the MetricNet module[98mrx 2023] is introduced to estimate per-pixel reliability maps for the forward and backward flows. Given $I_0, I_1, f_{0→1}, f_{1→0}$, MetricNet computes the initial photometric differences:

$$M_0 = \|I_0 - \text{warp}(I_1, f_{0→1})\|_1, M_1 = \|I_1 - \text{warp}(I_0, f_{1→0})\|_1 \quad (1)$$

where the warping operation aligns one frame to the other using backward warping. It further incorporates forward-backward consistency checks[Meister et al. 2017] and concatenates image, flow, and occlusion information into a feature tensor X . This tensor passes through several residual convolutional layers to predict confidence maps:

$$\text{Metric}_0, \text{Metric}_1 = \tanh(\text{MetricNet}(X)) \times 10 \quad (2)$$

where Metric_0 and Metric_1 represent the soft visibility weights for the two input frames. These maps are later used as blending coefficients in the soft warping step.

For a desired interpolation timestep $t \in (0, 1)$, the intermediate flow fields from each input frame to the target frame are linearly scaled:

$$f_{0→t} = t \cdot f_{0→1}, f_{1→t} = (1 - t) \cdot f_{1→0} \quad (3)$$

Correspondingly, the weighted confidence maps are interpolated as:

$$\text{Metric}_{0→t} = t \cdot \text{Metric}_0, \text{Metric}_{1→t} = (1 - t) \cdot \text{Metric}_1 \quad (4)$$

These scaled flows and weights provide motion and reliability priors for synthesizing the intermediate representation.

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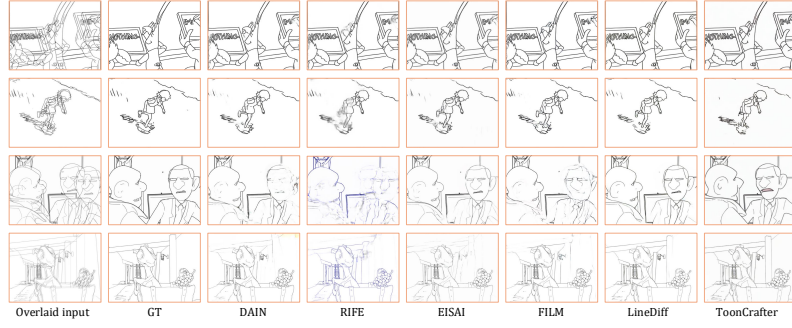


Figure 2: Qualitative comparison between the proposed LineDiff and the state-of-the-art interpolation methods.

LineDiff employs a soft warping operator, implemented via the SoftSplat layer[Niklaus and Liu 2020], to handle multiple pixels mapping to the same target location due to motion overlap. Unlike standard bilinear warping, SoftSplat performs forward splatting using the confidence weights as importance factors:

$$\begin{aligned}\hat{I}_{0 \rightarrow t} &= \text{SoftSplat}(I_0, f_{0 \rightarrow t}, \text{Metric}_{0 \rightarrow t}), \\ \hat{I}_{1 \rightarrow t} &= \text{SoftSplat}(I_1, f_{1 \rightarrow t}, \text{Metric}_{1 \rightarrow t})\end{aligned}\quad (5)$$

This operation effectively blends pixels from each frame according to both their motion and visibility likelihood, preventing ghosting artifacts in occluded regions.

In the meantime, to capture rich structural and semantic information, each input frame passes through the FeatureNet encoder, a lightweight three-level convolutional hierarchy:

$$\phi_i^k = \text{FeatureNet}(I_i), k = 1, 2, 3, i = \{0, 1\} \quad (6)$$

where ϕ^1, ϕ^2, ϕ^3 correspond to features at progressively coarser resolutions (1/2, 1/4, and 1/8 scales). Each set of features is warped using the same flow and confidence fields (appropriately downsampled), producing multi-scale warped feature pairs:

$$\begin{aligned}\hat{\phi}_{0 \rightarrow t}^k &= \text{SoftSplat}(\phi_0^k, f_{0 \rightarrow t}^k, \text{Metric}_{0 \rightarrow t}^k), \\ \hat{\phi}_{1 \rightarrow t}^k &= \text{SoftSplat}(\phi_1^k, f_{1 \rightarrow t}^k, \text{Metric}_{1 \rightarrow t}^k)\end{aligned}\quad (7)$$

These warped features serve as inputs to the fusion network, enabling context-aware reconstruction that integrates motion, structure, and appearance cues across scales.

The FusionNet module fuses the warped images and features into a coherent, high-quality interpolated frame. It takes as input the concatenated tensors:

$$\begin{aligned}X_{img} &= [I_0, \hat{I}_{0 \rightarrow t}, \hat{I}_{1 \rightarrow t}, I_1] \\ X_{feat}^k &= [\hat{\phi}_{0 \rightarrow t}^k, \hat{\phi}_{1 \rightarrow t}^k], k = 1, 2, 3\end{aligned}\quad (8)$$

FusionNet consists of a three-level grid topology with residual, downsample, and upsample connections linking horizontal and vertical streams. At each scale, residual convolutional blocks refine features, while skip connections propagate both fine details and contextual information between resolutions. The final output is reconstructed using a pixel shuffle block for spatial upsampling:

$$\hat{I}_t = \text{FusionNet}(X_{img}, X_{feat}^1, X_{feat}^2, X_{feat}^3) \quad (9)$$

Table 1: AnimeRun[Siyao et al. 2022] Validation Set Overview.

	LPIPS↓	CD↓	PSNR↑	SSIM↑
DAIN[Bao et al. 2019]	9.901	22.592	22.346	<u>0.842</u>
RIFE[Huang et al. 2020]	16.415	32.849	20.559	0.816
EISAI[Chen and Zwicker 2021]	10.813	39.386	21.606	0.838
FILM[Reda et al. 2022]	<u>9.821</u>	<u>19.182</u>	<u>22.314</u>	0.840
ToonCrafter[Xing et al. 2024]	20.902	705.138	17.307	0.715
LineDiff	9.457	18.656	21.264	0.843

3 Experiment

We conduct a quantitative comparison with other SOTA models on AnimeRun’s validation set [Siyao et al. 2022], with results presented in Table ?? . The best and runner-up values are **bold** and underlined, respectively. We employ metrics as PSNR, SSIM, LPIPS, and chamfer distance (CD) to quantify perceptual quality. The results are evaluated using each model’s official weights.

Refer to Figure 2, we also perform qualitative analysis on STD-12K [Li et al. 2021; Shen et al. 2024] and AnimeRun dataset [Siyao et al. 2022]. It can be observed that LineDiff is capable of generating high-quality in-betweens, and the results produced by other methods generally have different types of artifacts, such as blurriness and distortions.

4 Conclusion

This work explores the possibility of incorporating a generative optical flow network into a traditional warp-and-synthesis frame interpolation network. The proposed LineDiff framework offers a promising direction for bridging deterministic and generative paradigms in animation frame interpolation. By embedding generative reasoning into motion estimation while retaining a stable rendering backend, this framework paves the way for more controllable and reliable generative systems in animation production. The source code is available at <https://github.com/Soooda/LineDiff>.

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